

MCM/ICM2025-B

Managing Sustainable Tourism

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2024APMCM-A 一等奖，2025策联杯-B COPT优化应用奖，2025CUMCM-C 上海赛区二等奖

2025年11月23日

饮水思源 · 爱国荣校



真题介绍

问题分析

论文精读

总结回顾

背景引入：过度旅游业引发焦虑

美国阿拉斯加州**朱诺市（Juneau）**人口约3万，2023年创下160万人次邮轮游客的新纪录。虽然这些游客为城市带来了约3.75亿美元的可观收入，但也引发了过度拥挤的问题，迫使市政府不得不限制游客数量。

朱诺主要景点之一的**门登霍尔冰川（the Mendenhall Glacier）**，正因过度旅游导致的气温升高而持续消退。自2007年以来，冰川消退面积相当于八个足球场，这让许多当地居民担忧游客和相关收入终将随着冰川消失。值得庆幸的是，朱诺还有**观鲸、雨林**等其他景点，只要制定并实施**可持续旅游计划**，仍可保持其旅游目的地地位。

近期报告指出，旅游业存在**隐性成本（the hidden costs of tourism）**，亟需核算和管理这些成本，以保护自然资源与文化遗产，构建**可持续旅游产业**。——这些隐性成本包括对当地**基础设施的压力**，以及环境敏感区域旅游目的地**碳足迹的整体增加**。此外，过度旅游增大了当地**居民生活压力**（由于住房供应不足、成本上涨、过度拥挤、游客喧闹）。





真题介绍

问题分析

论文精读

总结回顾

问题综述： 可持续旅游业系统建模

- **¹Build a model for a sustainable tourism industry in Juneau**, Alaska. You may want to consider factors such as the number of visitors, overall revenue, and measures enacted to stabilize tourism. State clearly **²which factors you are optimizing, and which factors serve as constraints**. Include a plan for expenditures from any additional revenue and show how these expenditures feed back into your model to promote sustainable tourism. **³Include a sensitivity analysis and discuss which factors are most important**.
- **⁴Demonstrate how your model could be adapted to another tourist destination impacted by overtourism**. How does the choice of location effect which measures will be most important? How might you use your model to promote attractions and/or locations that have fewer tourists to develop a better balance?
- Write a **⁵one-page memo to the tourist council of Juneau** outlining your predictions, the effects of various measures, and your advice on how to optimize outcomes.

Task 1: **【机理机制建模】** 为阿拉斯加朱诺市构建可持续旅游产业模型。

Task 2: **【优化建模求解】** 明确界定优化因素与限制因素，制定额外收入的支出方案，并说明这些支出如何反哺模型以促进可持续旅游发展。

Task 3: **【敏感性分析】** 重点探讨关键影响因素。

Task 4: **【模型迁移应用】** 展示如何模型应用于其他受过度旅游影响的旅游目的地。
选址如何影响哪些指标最为重要？如何利用该模型推广游客较少的景点或地区，以实现更好的平衡？

Task 5: **【非技术性文本】** 为朱诺市旅游局撰写一份一页备忘录。
需要详细阐述预测分析、各项措施的预期效果，并提出优化成果的具体建议。



问题分析

论文精读

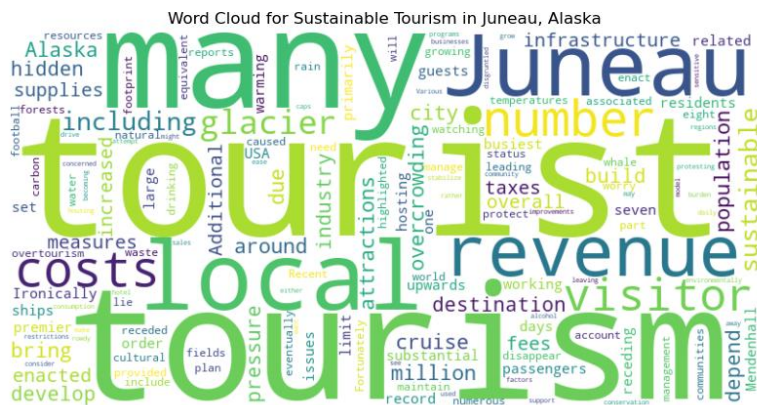
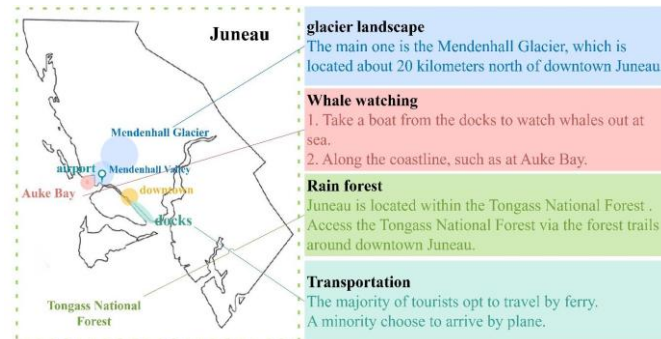
总结回顾

任务 1: Build a model for a sustainable tourism industry in Juneau, Alaska. You may want to consider factors such as the number of visitors, overall revenue, and measures enacted to stabilize tourism.

——考虑多种因素，为阿拉斯加朱诺市构建可持续旅游产业模型。

思考：

- 为了构建起可持续旅游业模型，需要选用哪些因素？
- 这些因素应该以何种数学形式呈现？
- 旅游业政策如何作用于整个产业？←-----



Additional revenue from taxes have been used to support conservation, make improvements in infrastructure, and develop community programs. While many locals that depend on tourism are concerned that additional fees might drive tourists away and would rather see the numbers, and their businesses, grow, many other locals are becoming disgruntled and either leaving or protesting against tourists.



真题介绍

问题分析

论文精读

总结回顾

任务一：构建可持续旅游产业模型

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思考：

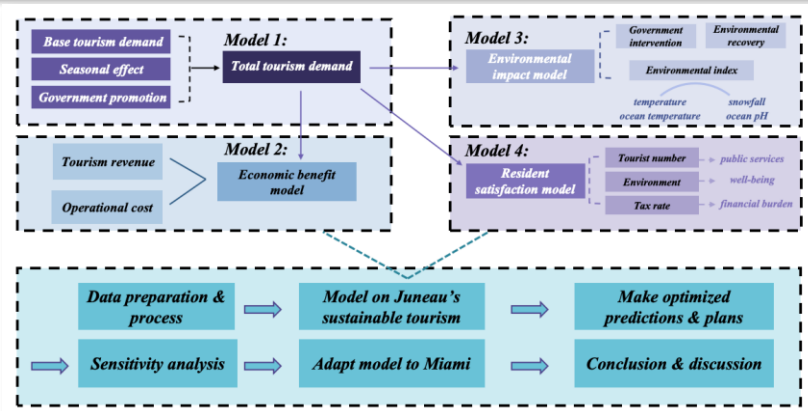
- 选用哪些因素？

city working to limit the number of guests. Ironically, Mendenhall Glacier, one of the premier attractions in Juneau, has been receding, primarily due to warming temperatures caused, in part, by overtourism. The glacier has receded the equivalent of eight football fields since 2007, leading many locals to worry that the tourists and associated revenue will eventually disappear with the glacier.^[3] Fortunately, Juneau has other attractions, including whale watching and rain forests, and can maintain their status as a tourist destination, provided they can develop and enact a plan for sustainable tourism.

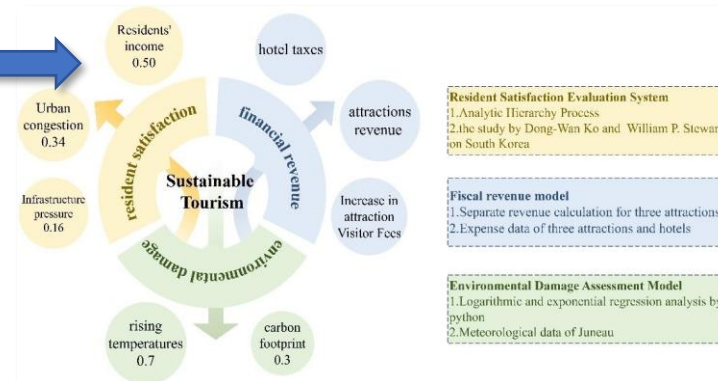
Recent reports^[4] have highlighted the hidden costs of tourism and the growing need to account for and manage these costs to protect natural and cultural resources and build a sustainable tourism industry on which numerous communities around the world depend. These hidden costs include pressure on local infrastructure including drinking water supplies, waste management, and an overall increased carbon footprint in tourist destinations, many of which lie in environmentally sensitive regions. Local populations are also under pressure due to housing supplies and costs, overcrowding, and rowdy tourists. Various measures have been enacted to attempt to ease the burden, including increased hotel taxes, visitor fees, caps on the number of daily visitors, and restrictions on alcohol sales and consumption. Additional revenue from taxes have been used to support conservation, make improvements in infrastructure, and develop community programs. While many locals that depend on tourism are concerned that additional fees might drive tourists away and would rather see the numbers, and their businesses, grow, many other locals are becoming disgruntled and either leaving or protesting against tourists.

- Build a model for a sustainable tourism industry in Juneau, Alaska. You may want to consider factors such as the number of visitors, overall revenue, and measures enacted to stabilize tourism. State clearly which factors you are optimizing, and which factors serve

从题干和产业经济知识中获得信息



多个子模型统合：【客流】+【经济-环境-社会】



强调多方面因素的相互关联性：【社会*经济*环境】



真题介绍

问题分析

论文精读

总结回顾

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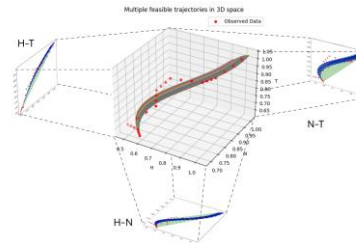
——考虑多种因素，为阿拉斯加朱诺市构建可持续旅游产业模型。

思考：

- 这些因素应该以何种数学形式呈现？

例：受Lotka-Volterra模型、Logistic人口模型和SIS模型启发，用常微分方程描绘动态关联性

$$\begin{aligned}\frac{dH_{\text{Juneau}}}{dt} &= k_1 H_{\text{Juneau}} + k_2 H_{\text{Juneau}} N_{\text{Juneau}} + k_3 T_{\text{Juneau}} H_{\text{Juneau}} + \epsilon_{H_{\text{Juneau}}}, \\ \frac{dN_{\text{Juneau}}}{dt} &= r_1 N_{\text{Juneau}} \left(1 - \frac{N_{\text{Juneau}}}{K}\right) + k_4 H_{\text{Juneau}} N_{\text{Juneau}} + \epsilon_{N_{\text{Juneau}}}, \\ \frac{dT_{\text{Juneau}}}{dt} &= k_5 T_{\text{Juneau}} + k_6 N_{\text{Juneau}} + \epsilon_{T_{\text{Juneau}}}.\end{aligned}$$



又例：从公共财政学中的社会效用模型获得灵感，用对数形式描绘社会效益的边际效应

$$(\text{社会层面}) S_t = [a_0 - \alpha_{soc} \cdot \ln(R_t \theta_{t,soc})] N_t + b_0$$

但是简简单单的线性关系绝对不是“低级”、“不好”的。

- 旅游业政策如何作用于整个产业？

可以融入因素性的式子，也可以将前馈的政策、旅游业收入、后馈的政策视作Markov性的过程...



真题介绍

问题分析

论文精读

总结回顾

任务二、三：优化模型与敏感性分析

任务 2: State clearly which factors you are optimizing, and which factors serve as constraints. Include a plan for expenditures from any additional revenue and show how these expenditures feed back into your model to promote sustainable tourism.

——建立优化模型并求解分析。

任务 3: Include a **sensitivity analysis** and discuss which factors are most important.

——对于求解结果进行敏感性分析。

思考：

- 本题的优化问题情境是怎么样的？
- 这样的优化问题适合怎么样的求解方法？
- 什么是敏感性分析？
- 这里怎么做敏感性分析？



真题介绍

问题分析

论文精读

总结回顾

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思考：

• 本题的优化问题情境是怎么样的？适合怎么样的求解方法？

- 【多个子模型】多目标/【高度耦合】处理后的单目标
- 大部分论文在可持续旅游业建模时都不可能避开非线性性
- 约束体系总体上较为简单

• 启发式算法——SA算法、GA算法、PSO算法、NSGA-II...

- 例：使用NSGA-II算法求解多目标优化问题，获得其Pareto前沿
- （但也有论文使用DP（动态规划）解决问题）

5.2 NSGA-II for Multi-objective Optimization

The model employs Non-dominated Sorting Genetic Algorithms (NSGA II) for multi-objective optimization, seeking the optimal balance among **financial revenue**, **environmental damage**, and **residents' satisfaction**. NSGA-II is an improved multi-objective genetic algorithm. It deals with multi-objective optimization problems by stratifying population individuals through a non-dominated sorting mechanism. The algorithm flow chart can be viewed in Figure 8.

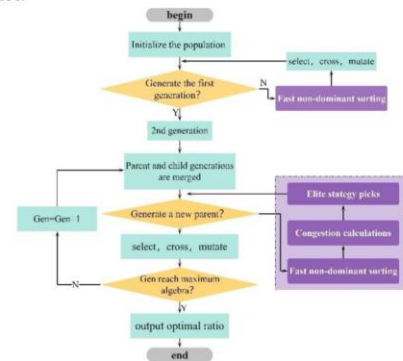


Figure 8: NSGA II Algorithm Flowchart

After solving with Python, a Pareto front containing 255 solutions was obtained.

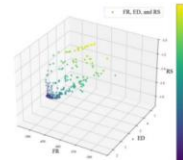


Figure 9: Pareto Front



真题介绍

问题分析

论文精读

总结回顾

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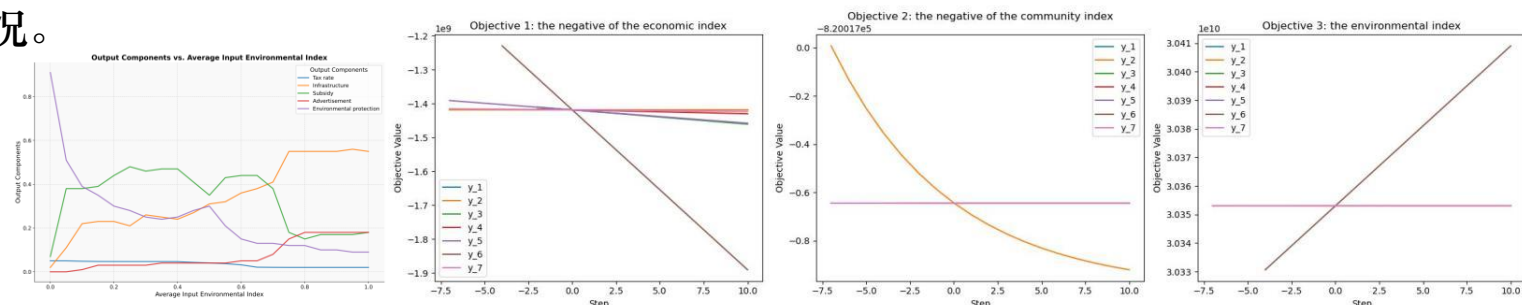
思考：

- 什么是敏感性分析？这里怎么做敏感性分析？

敏感性分析衡量的是：当模型的输入（参数、数据）发生小到大的变化时，输出（最优解下的旅游业政策）如何变化。

——哪些输入更加具有“决定性”（目标函数对其更敏感）？

衡量因子的局部敏感性，可以使用OAT法（一次只变一个，One-at-a-time）加扰动，考察目标函数的变化情况。





真题介绍

问题分析

论文精读

总结回顾

任务四、五：模型迁移与非技术性文本

任务 4: Demonstrate how your model could be adapted to another tourist destination impacted by overtourism. How does the choice of **location** effect which measures will be most important? How might you use your model to promote attractions and/or locations that **have fewer tourists** to develop a better balance?

——将可持续旅游业模型迁移到其他存在过度旅游的旅游城市。

任务 5: Write a one-page memo to the tourist council of Juneau outlining your predictions, the effects of various measures, and your advice on how to optimize outcomes.

——为朱诺市旅游局写一份一页备忘录。

思考：

- “其他存在过度旅游的旅游城市”选什么比较好？怎么迁移？
- 如何写非技术性文本？



真题介绍

问题分析

论文精读

总结回顾

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思考：

- “其他存在过度旅游的旅游城市”选什么比较好？怎么迁移？

必选项目：fewer tourists（如Reine）

其他选择：具有人文景观的景点（如Machu Picchu），地理条件迥异的景点（如Great Barrier Reef）

迁移方式：重新找数据来获得参数、修改部分变量/因子的含义（如冰山面积），再次敏感性分析

Recent reports^[4] have highlighted the **hidden costs** of tourism and the growing need to account for and manage these costs to protect natural and cultural resources and build a sustainable tourism industry on which numerous communities around the world depend. These hidden costs



真题介绍

问题分析

论文精读

总结回顾

任务四、五：模型迁移与非技术性文本

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——为朱诺市旅游局写一份一页备忘录。

思考：

• 如何写非技术性文本？

- **读者意识：**以“决策者为中心”写作——想象收件人是一位很忙的官员——他们要结论、要点、可执行的建议、以及能支持决策的关键风险/收益，不要让他们去“读模型”。
- **结论先行：**先一句话概述最重要的结论或建议，再展开支持点。
- **结构清晰：**关键预测 + 措施（短）+ 直接预期效果（**量化**或定性）。
- **回避技术性阐述：**回避专业性、数学性过强的表述，使用礼貌、客观、易读、平直的语言。



真题介绍

问题分析

论文精读

总结回顾

思路梳理及框架对比

核心思路：将问题拆解为经济、环境、社会等方面，对于各个方面分别建模，合成单目标综合优化指标，或多目标动态系统，采用规划类方法加以求解，对于结果进行敏感性分析和跨地区的迁移运用

	O 奖	F 奖	M 奖	H 奖	S 奖
问题分解	环境、经济、社会	环境、旅游收入(经济)、游客与居民幸福指数(社会)	冰川保护(环境)、收入(经济)、幸福指数(社会)	环境、经济、社会	环境、经济、社会
核心方法	System Dynamics, L-BFGS-B算法	Genetic Algorithm(遗传算法)	AHP(层次分析法) 牛顿二分法	AHP, 梯度下降, 递归分析	AHP, 遗传算法
目标函数类型	多目标动态平衡	单目标	单目标	单目标	单目标
函数类型	线性非线性函数, 微分方程, logistic函数	柯布-道格拉斯函数, 线性非线性函数	Sigmoid函数, 线性非线性函数	Logistic函数, 线性非线性函数	Logistic函数, 线性非线性函数
敏感性分析	历史拟合验证 + 长期仿真曲线收敛	Hessian 矩阵验证最大值点性质	简单敏感性分析	简单敏感性分析	简单敏感性分析
跨地区迁移	旅游业发展过度 + 缺乏地区	理论分析, 未引入实例	仍在分析Juneau 市的情形	旅游业发展过度 + 当地冷门地区	未提及



真题介绍

问题分析

论文精读

总结回顾

建模亮点

任务 4: Demonstrate how your model could be adapted to another tourist destination impacted by overtourism. How does the choice of **location** effect which measures will be most important? How might you use your model to promote attractions and/or locations that **have fewer tourists** to develop a better balance?

——将可持续旅游业模型迁移到其他存在过度旅游的旅游城市。

O奖论文不仅把模型扩展到“过度旅游”的旅游城市，还扩展了思路，把模型迁移到欠发达的旅游城市，探查本模型有什么方法去帮助发展

8.3 Promoting Undervisited Destinations via Dynamic Models

Our model can promote less-visited attractions by leveraging its feedback mechanisms and policy-adjustment capabilities. Through scenario simulations, we can identify areas with lower tourist volumes and high development potential. By adjusting key parameters such as infrastructure investment (e.g., increasing θ_{infra}), tax rates, and public service allocation, we can enhance the appeal of these locations while avoiding overcrowding. Specifically, targeted infrastructure improvements can enhance accessibility and visitor experience, while environmental conservation investments ensure sustainable growth. Simultaneously, promotional efforts focusing on the unique cultural and ecological features of these areas can help shift tourist demand, achieving a balanced distribution of visitors. Additionally, the model allows continuous monitoring of resident satisfaction and environmental health to ensure the long-term success of these measures.



真题介绍

问题分析

论文精读

总结回顾

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——建立优化模型并求解分析。

O奖论文没有简单把多目标合成单目标，而是在一个系统的、动态的体系中综合地对参数进行调整，从而实现长期的较优水平：

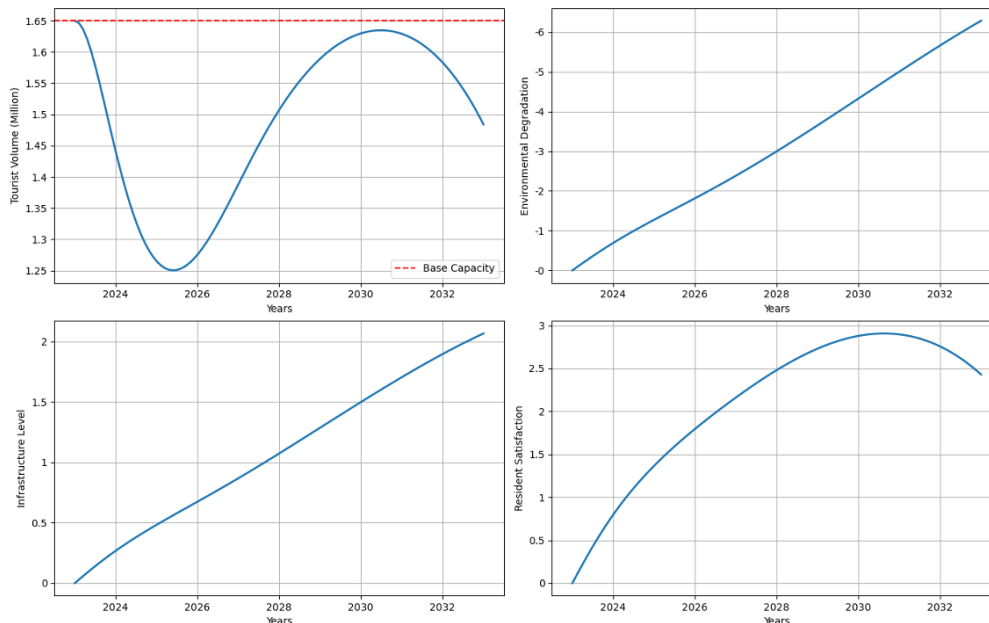


Figure 8: Visualization of program output results

同时，也没有僵化地求解政策手段，而是拆分阶段，预测未来动向，有变化地求解，将目光放得很长远

- Short Term (1-3 years): 通过临时提高税率实施价格调控，以抑制游客过载并缓解紧急的生态与拥堵压力
- Mid Term (3-5 years): 随着客流回落，将重心转向基础设施与环保的高投入，旨在提升居民满意度并扩充生态容量
- Long Term (5+ years): 建立环境、经济与社会的正反馈循环，通过动态监测与持续高投入，实现游客适度增长与低环境退化的可持续共存



真题介绍

问题分析

论文精读

总结回顾

表达式简明化

- 对于建模过程，不是说式子越复杂越好
- 好的建模过程，表达式要具有形式上的美观性和内容上的可解释性
美观性：变量名的选取要清晰，可以用大括号来划分不同项表示的含义
可解释性：从情境本身出发，从文献中找到依据
- 如果没有理论支撑，那么优先选择简洁的式子（如线性表达式）

$$\begin{aligned}\frac{dx}{dt} &= \underbrace{\alpha_0(1 + \lambda_I I)}_{\text{Infrastructure boost}} \underbrace{(1 + \xi_S S)}_{\text{Satisfaction effect}} x - \underbrace{\beta_0(1 + \zeta_E E)}_{\text{Environmental pressure}} x^2 \left(\frac{x}{K}\right) \\ \frac{dI}{dt} &= \underbrace{\phi_{\text{infra}} \theta_{\text{infra}} T}_{\text{Investment flow}} - \underbrace{\delta_I I}_{\text{Depreciation rate}} \\ \frac{dE}{dt} &= \underbrace{\eta x(1 + \gamma \tau)}_{\text{Tourism pressure}} - \underbrace{\delta_E E}_{\text{Natural recovery}} - \underbrace{\phi_{\text{env}} \theta_{\text{env}} T}_{\text{restorative impact of environmental investments}} \\ \frac{dS}{dt} &= \underbrace{\mu_S(I + \epsilon \theta_{\text{pub}} T)}_{\text{Public service}} - \underbrace{\kappa_S \cdot x}_{\text{Congestion cost}} - \underbrace{\nu_S E}_{\text{Environmental degradation}} \\ T &= \underbrace{\tau x}_{\text{Tax revenue}} + \underbrace{f_{\text{other}} x}_{\text{Fixed fees}}\end{aligned}$$

列式清晰，有直观的注解
形式简洁，没有复杂的关系

Table 2: Initial Parameter Values and Data Sources

Symbol	Value	Data Source
E_0	0.1	CBJ Cruise Impacts 2023 Report[6]
x_0	165×10^6	CBJ Cruise Impacts 2023 Report[6]
S_0	0.7	CBJ Cruise Impacts 2023 Report[6]
I_0	0.4	CBJ Cruise Impacts 2023 Report[6]

Table 3: Parameter Values and Data Sources[2][3][4][5][6]

Equation	Symbol	Value	Data Source
(1): Tourist Dynamics	K	1,650,000	2023 cruise passenger volume (CBJ-Tourism-Survey-2023-Report, Page 9)
	α_0	0.3	Historical data calibration
	λ_I	0.2	Transportation investment return rate
	ξ_S	0.48	Resident satisfaction survey
	β_0	0.55	Environmental carrying capacity research
	ζ_E	0.43	Emission monitoring data
	γ	0.05	Fiscal policy analysis
(2): Infrastructure Dynamics	ϕ_{infra}	0.35	Weighted data from transportation and infrastructure development
	θ_{infra}	0.35	Reduce traffic congestion (55%) + develop public transportation (42%)
	δ_I	0.02	Infrastructure depreciation studies
(3): Environmental Dynamics	η	0.43	Weighted average of cruise emissions (36%), whale-watching boats (47%), and aviation noise (43%)
	θ_{env}	0.25	Resident prioritization survey (restricting cruises 28%, 2022 data)
	δ_E	0.03	Ecosystem recovery studies
	ϕ_{env}	0.04	Efficiency of environmental protection investments
(4): Satisfaction Dynamics	μ_S	0.65	Normalized from high-priority projects (transportation, hotels, and conferences, 2022 data)
	κ_S	0.55	Prioritization of traffic congestion (55%, 2022 data)
	ν_S	0.42	Cruise emission impacts (42%, 2022 data)
(5): Tax System	τ	0.1	Fiscal policy and tax adjustment simulation
	f_{other}	50 USD	Average fixed fees per tourist (2023 data)

把变量的值汇总到一张表格中，
避免表达式本身出现冗长的数据



真题介绍

问题分析

论文精读

总结回顾

表达式对比

$$\begin{cases} \text{Inc}(x) = 461,514,817.486 + 221.212x \\ \text{Dep}(x) = -4,104,627.182 + 20.512x \end{cases}$$

$$I(x) = 20799.5726 + 32527.7670 \ln \left(1 + \frac{x}{1,105,788.56} \right)$$

$$U(x) = \frac{1}{1 + e^{-0.000000141156(x-7295029)}}.$$

$$I(x) = 3114.4063 + 71353.9833 \ln \left(1 + \frac{x}{14959337.7} \right)$$

$$Score_{GHI} = 1.931 - 0.314 \cdot e^{0.967 \cdot NV} - k_1 \cdot \ln \left(\frac{n}{NV} + e^{-1} \right)$$

$$Score_{revenue} = 0.926 \cdot NV - 0.6 + k_2 \cdot \ln \left(\frac{n}{NV} + e \right)$$

$$Score_{happiness} = -NV + 2.1263 + k_3 \cdot n$$

过多拟合数据代入，式子不作解释，意义不大

$$\frac{dx}{dt} = \underbrace{\alpha_0(1 + \lambda_I I)}_{\text{Infrastructure boost}} \underbrace{(1 + \xi_S S)}_{\text{Satisfaction effect}} x - \underbrace{\beta_0(1 + \zeta_E E)}_{\text{Environmental pressure}} x^2 \left(\frac{x}{K} \right)$$

$$\frac{dI}{dt} = \underbrace{\phi_{\text{infra}} \theta_{\text{infra}} T}_{\text{Investment flow}} - \underbrace{\delta_I I}_{\text{Depreciation rate}}$$

$$\frac{dE}{dt} = \underbrace{\eta x(1 + \gamma \tau)}_{\text{Tourism pressure}} - \underbrace{\delta_E E}_{\text{Natural recovery}} - \underbrace{\phi_{\text{env}} \theta_{\text{env}} T}_{\text{restorative impact of environmental investments}}$$

$$\frac{dS}{dt} = \underbrace{\mu_S(I + \epsilon \theta_{\text{pub}} T)}_{\text{Public service}} - \underbrace{\kappa_S \cdot x}_{\text{Congestion cost}} - \underbrace{\nu_S E}_{\text{Environmental degradation}}$$

$$T = \underbrace{\tau x}_{\text{Tax revenue}} + \underbrace{f_{\text{other}} x}_{\text{Fixed fees}}$$

不出现实际数据，全部使用变量代替，有清晰的含义解释



真题介绍

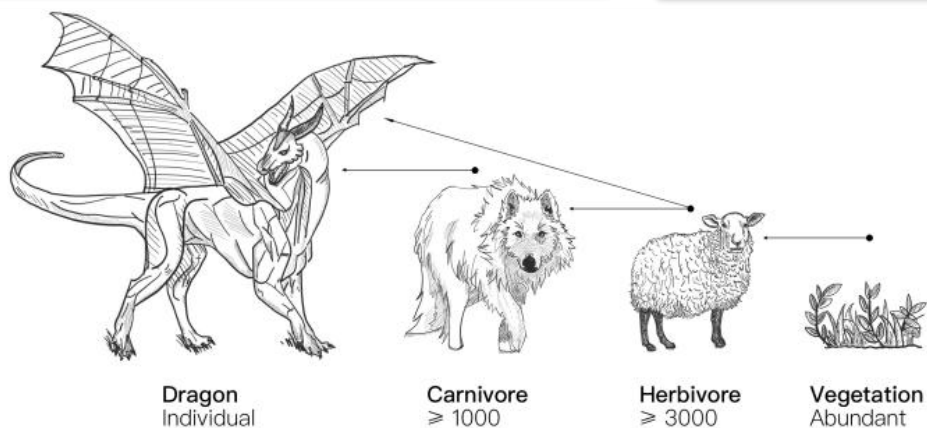
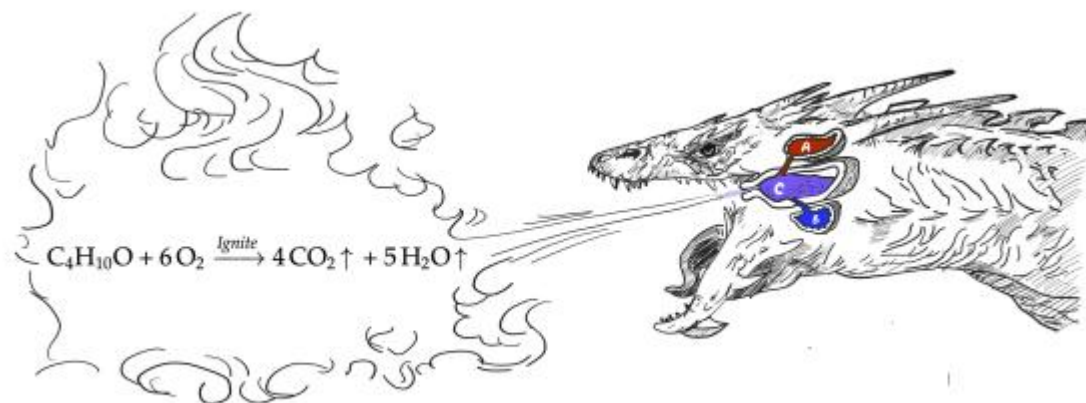
问题分析

论文精读

总结回顾

可视化分析

2019MCM-A: Game of Ecology



Northman
Weight: 65 kg
Energy Cons: 2000 cal/day
Ratio: x230

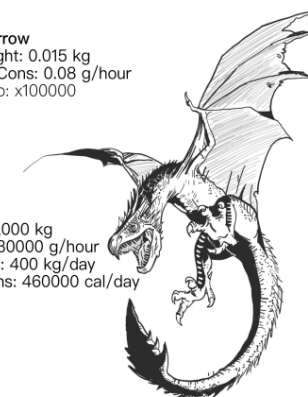


Sparrow
Weight: 0.015 kg
Fat Cons: 0.08 g/hour
Ratio: x100000

Direwolf
Weight: 80 kg
Meat Cons: 16 kg/day
Ratio: x188



Dragon
Weight: 15,000 kg
Fat Cons: 80000 g/hour
Meat Cons: 400 kg/day
Energy Cons: 460000 cal/day
Ratio: x1





真题介绍

问题分析

论文精读

总结回顾

可视化对比分析



0 奖



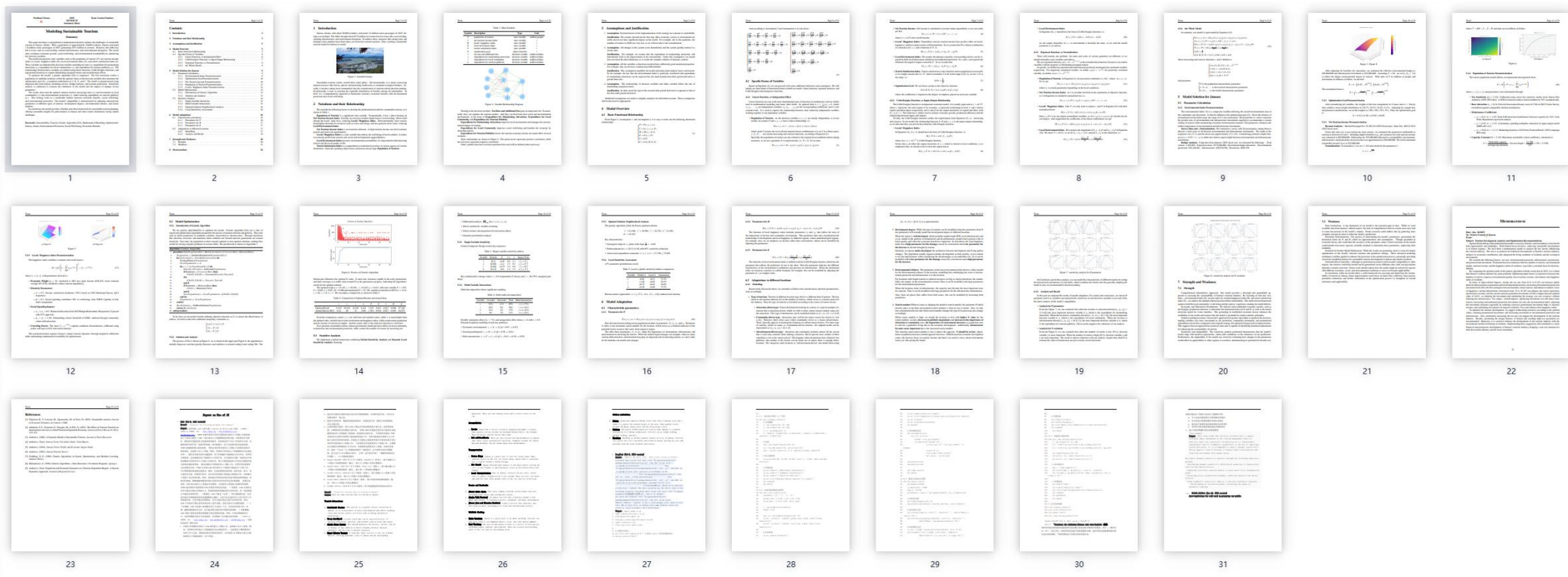
真题介绍

问题分析

论文精读

总结回顾

可视化对比分析



F 奖



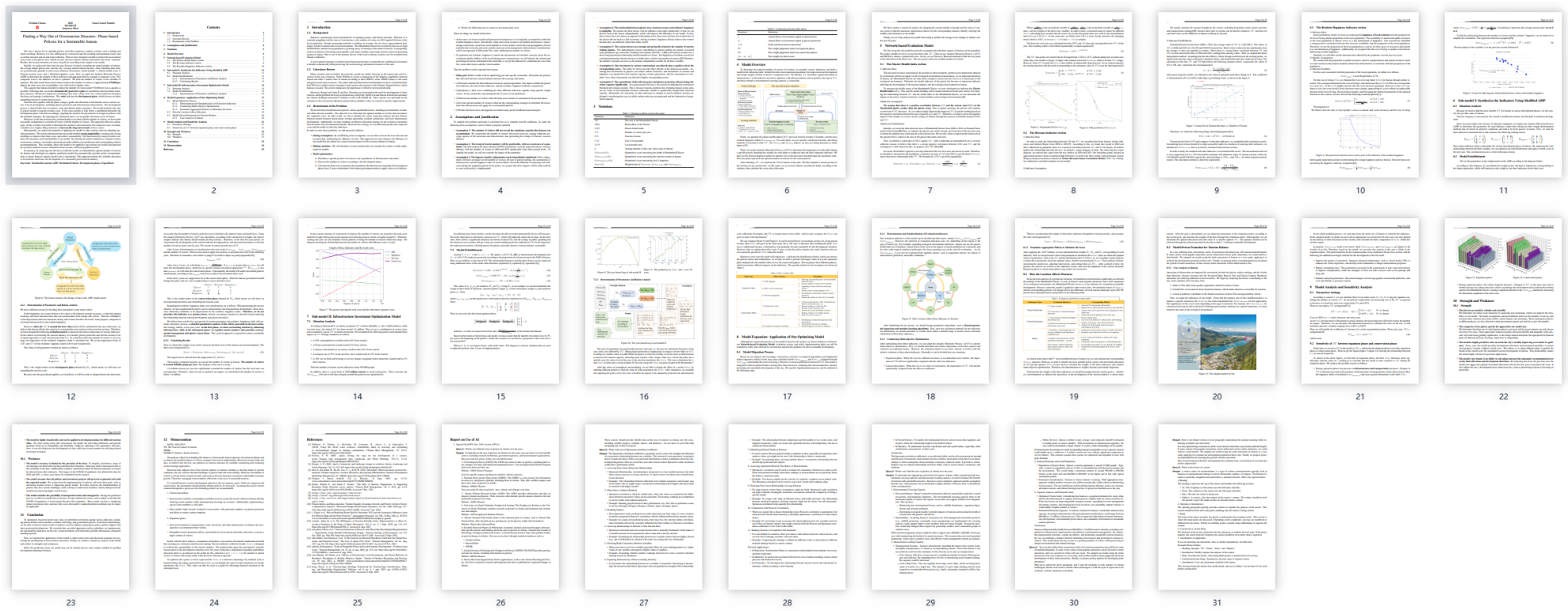
真题介绍

问题分析

论文精读

总结回顾

可视化对比分析



M 奖



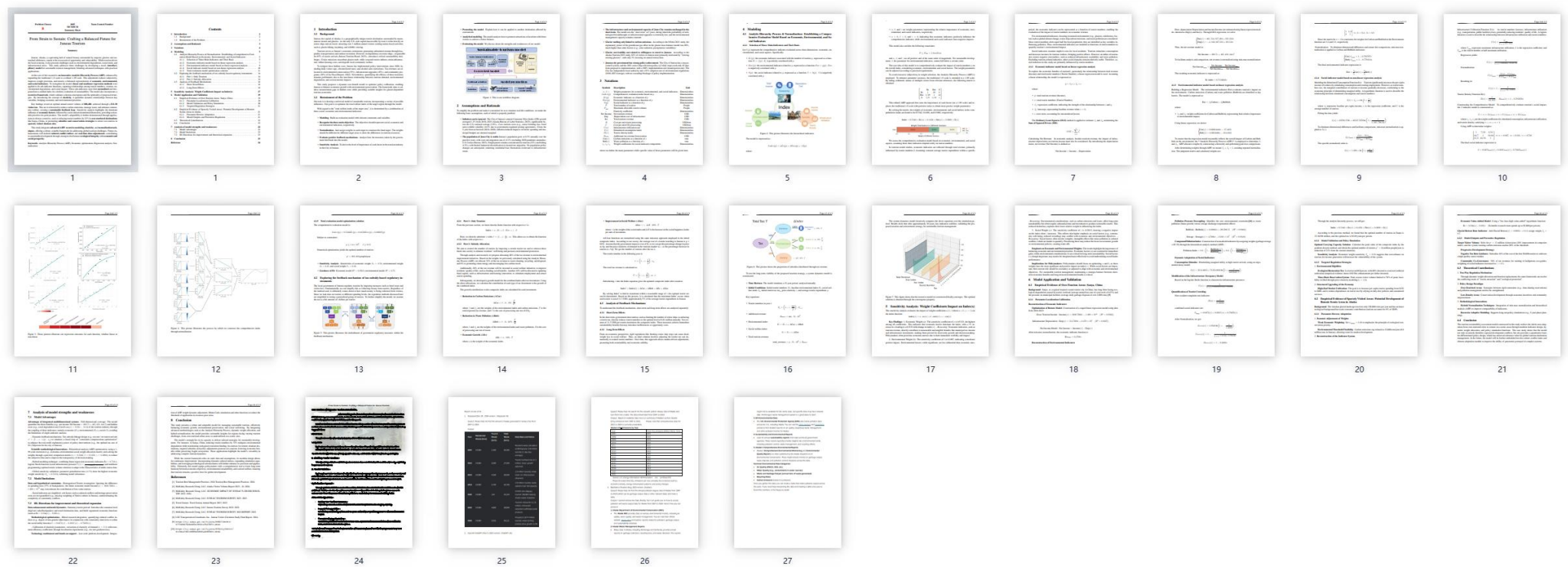
真题介绍

问题分析

论文精读

总结回顾

可视化对比分析



H 奖



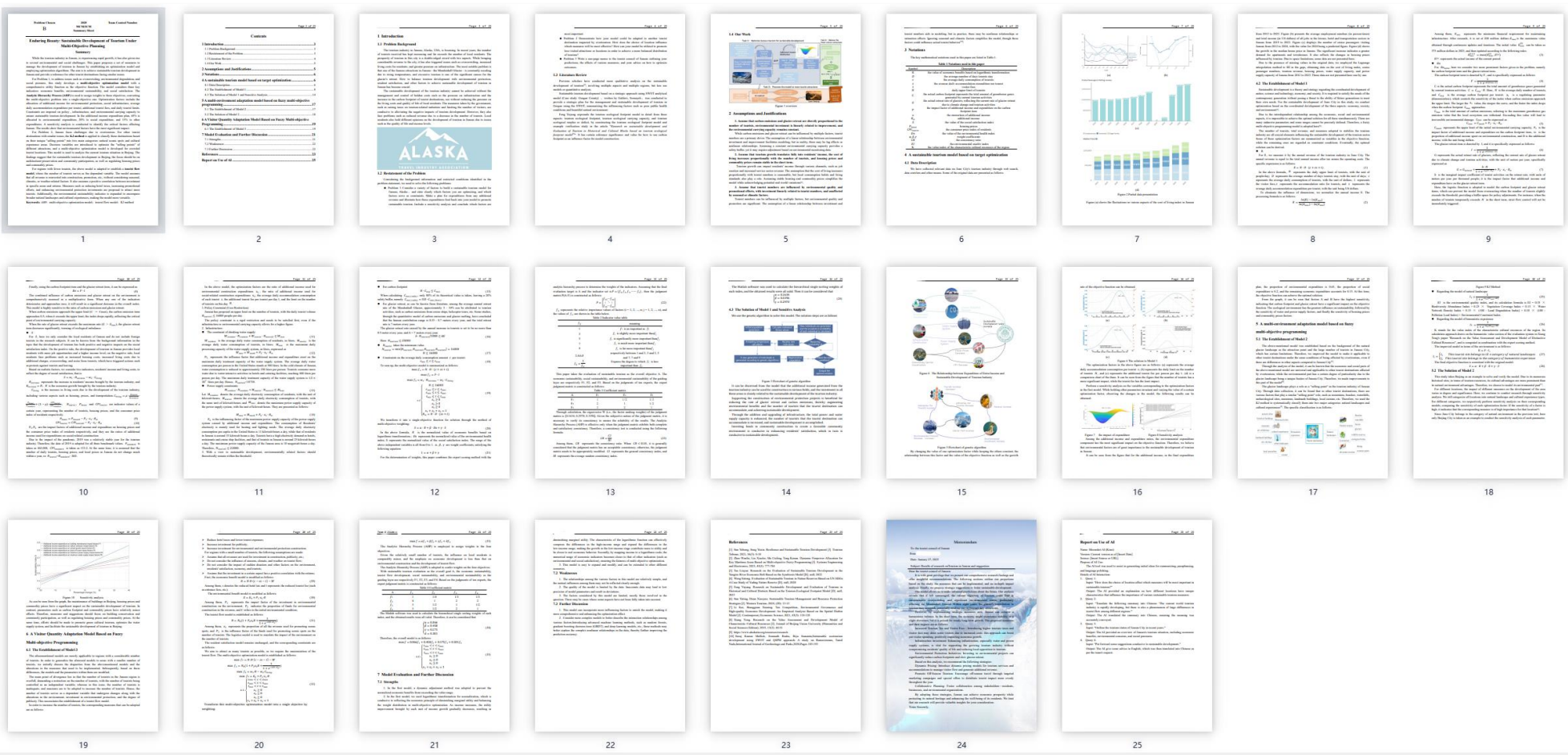
真题介绍

问题分析

论文精读

总结回顾

可视化分析



S 奖



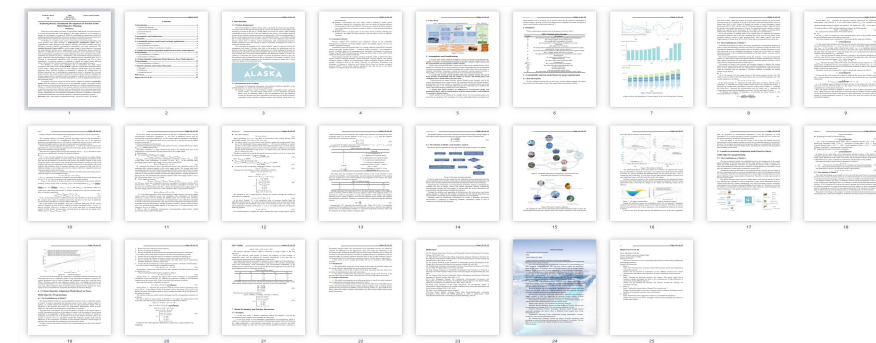
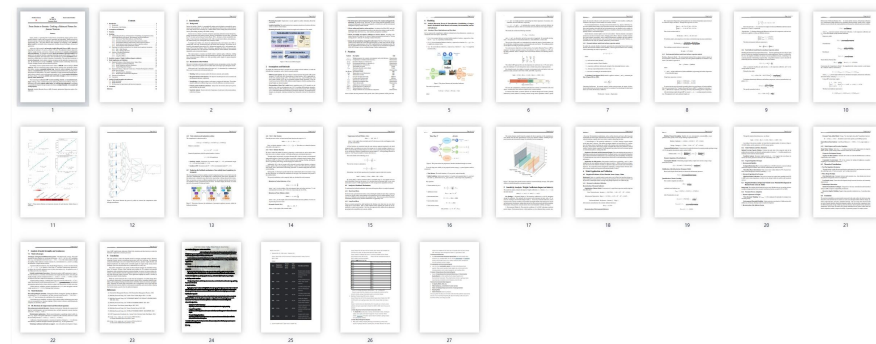
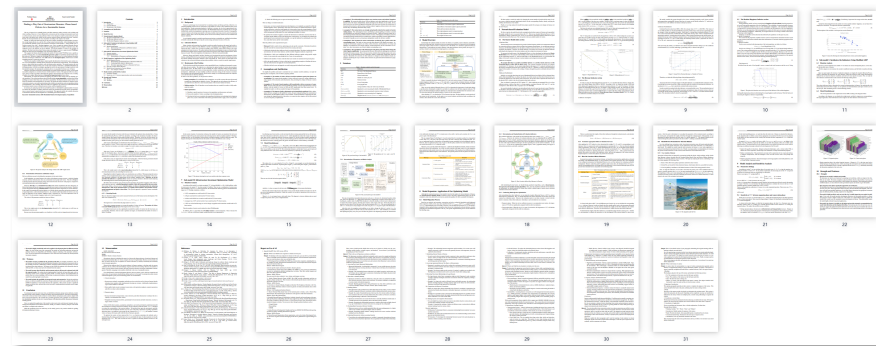
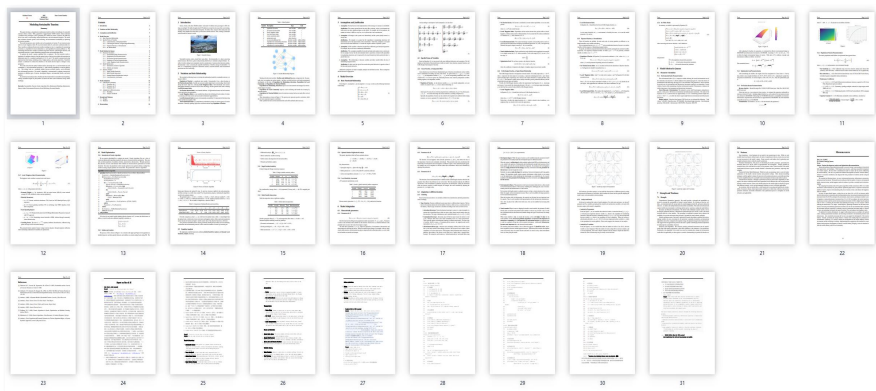
真题介绍

问题分析

论文精读

总结回顾

可视化分析





真题介绍

问题分析

论文精读

总结回顾

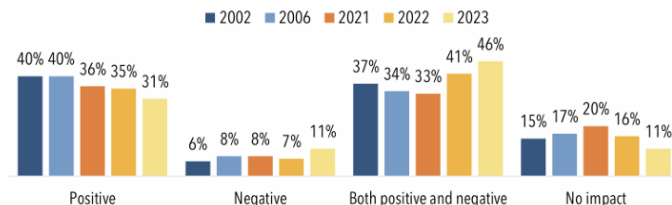
可视化重点总结

- 示意图与数据图兼而有之，示意图来阐释框架、思路；数据图来展示建模优化结果
- 重视可视化的多样性（框图，柱状图，函数图，三维函数图，热图.....）
- 可视化是论文的“气口”，能够调整论文的节奏，总结文段，辅助说理。在需要的时候及时出现(一整段的思路阐述后，展示完解题过程后，收集数据后.....)
- 配色美观淡雅，清晰简明



Figure 1: Juneau Tourism Impact Assessment

Comparison: Overall Impact of Tourism on Households, 2002, 2006, 2021, 2022, 2023



Notes: The 2021 survey referred to 2019 impacts. Excludes "don't know" and refused responses.

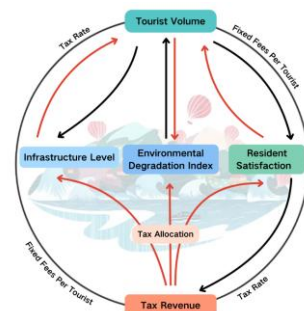


Figure 7: Dynamic Feedback Loops in Sustainable Tourism Systems

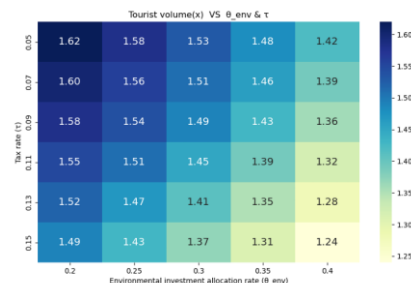


Figure 9: Tourist Volume VS Env. Investment Allocation Rate / Tax Rate

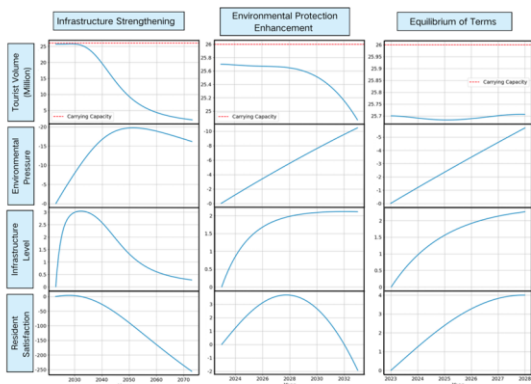


Figure 14: Visualization of Program Output Results for Sanya

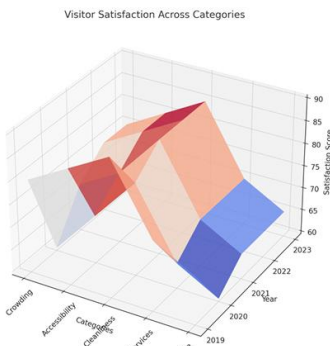


Figure 6: Visitor Satisfaction Across Categories[6]

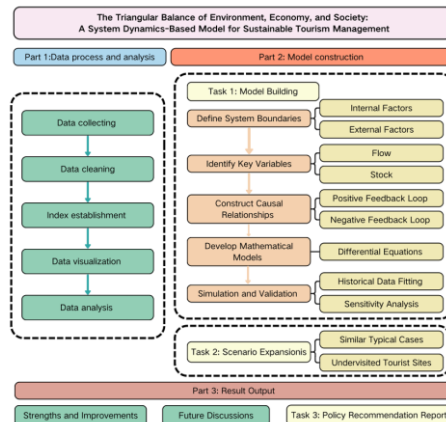


Figure 2: Our Work

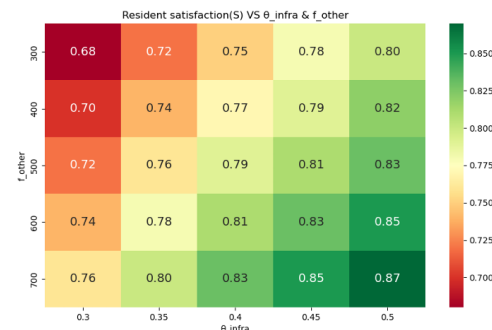


Figure 10: Resident Satisfaction VS Infra. Investment Allocation Rate & Fi



真题介绍

问题分析

论文精读

总结回顾

缺点分析

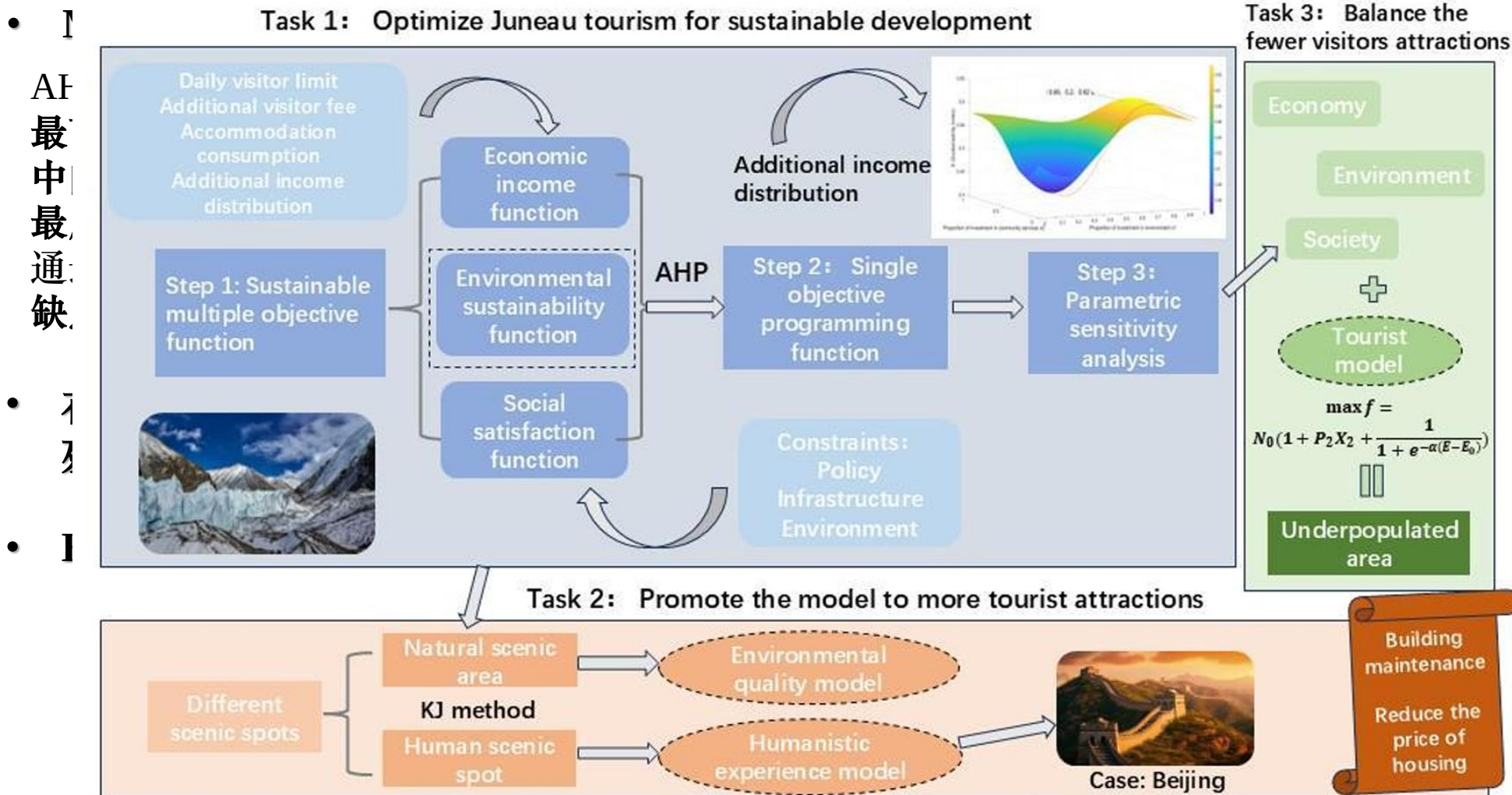


Figure 1 overview



真题介绍

问题分析

论文精读

总结回顾

缺点分析

- **M,H,S: 主观性强, 比较大比重地使用AHP这样依赖主观打分的机制方法**

AHP 的核心思想是把一个复杂的问题拆解成一个树状结构:

最高层 (目标层): 你想要达成的最终目标。

中间层 (准则层): 影响决策的各种因素。

最底层 (方案层): 你可供选择的备选方案。

通过两两比较这些因素, 计算出每个因素的权重, 最后算出每个方案的综合得分。

缺点: 主观性较强, 输入的判断矩阵完全依赖专家的主观打分, 如果打分偏颇, 结果也会偏颇。

- **在线性拟合后, 没有进行统计分析, 如计算决定系数 R^2 , 均方误差MSE, 残差 S_e , 导致逻辑未形成闭环, 模型建立缺乏依据**
- **H,S: 图片处理失误, 清晰度低, 降低了论文的专业性**

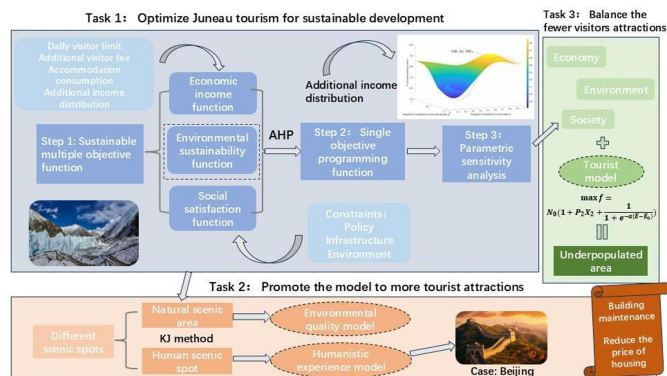


Figure 1 overview



真题介绍

问题分析

论文精读

总结回顾

缺点分析

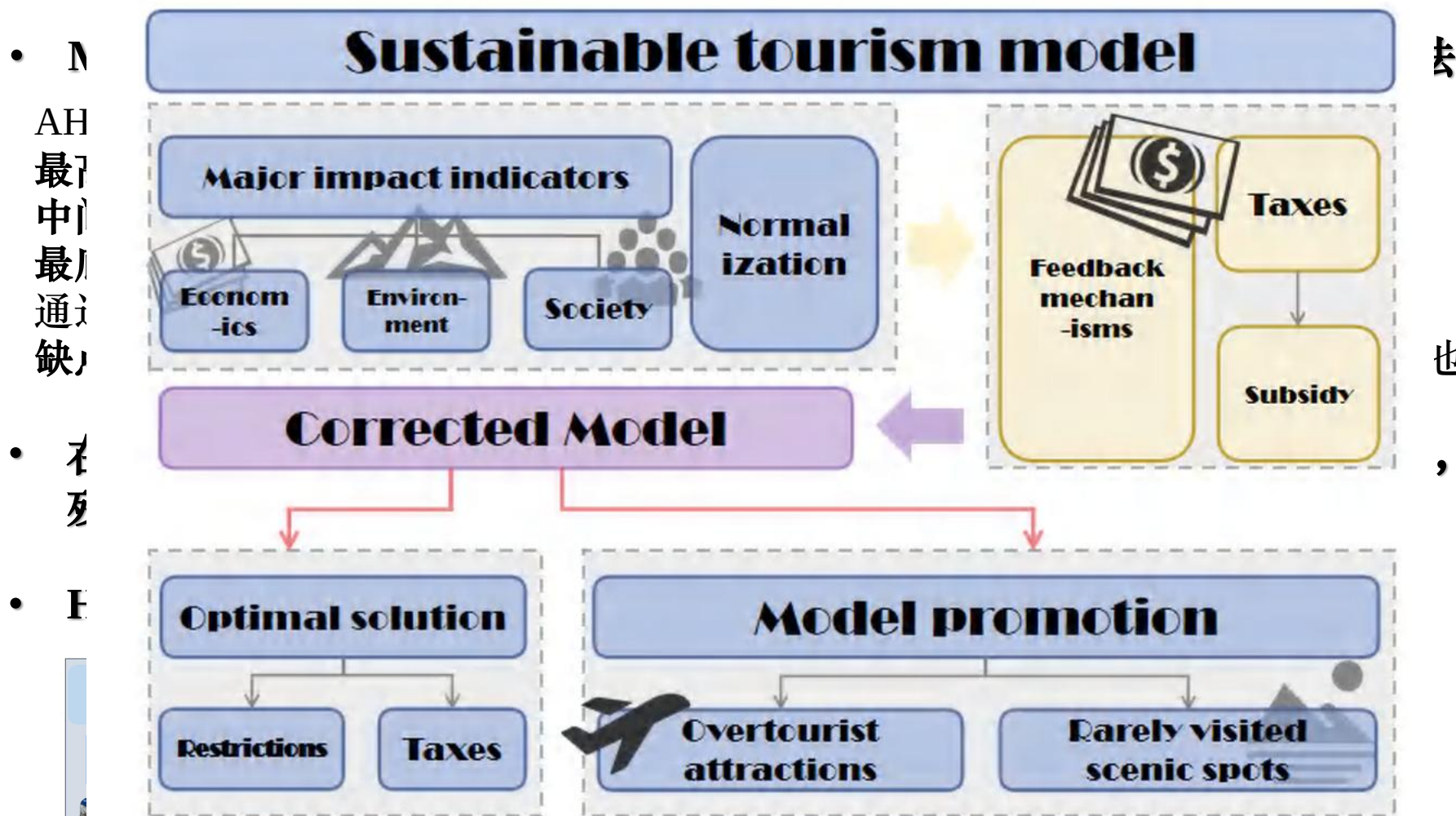


Figure 1: This is our workflow diagram.



真题介绍

问题分析

论文精读

总结回顾

缺点分析

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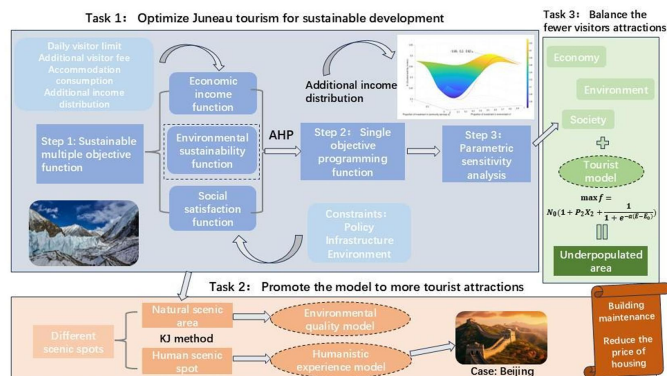


Figure 1 overview

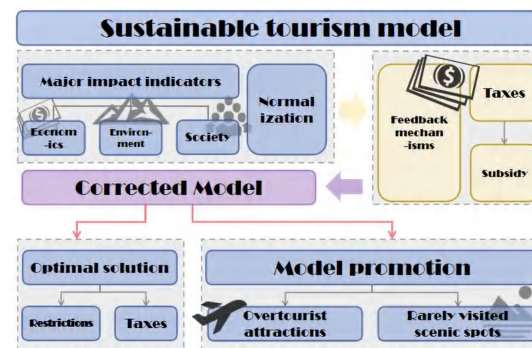


Figure 1: This is our workflow diagram.



真题介绍

问题分析

论文精读

总结回顾

总结与建议

- 本题属于数学建模中一道非常典型的题，建议初次接触数模的同学们可以多多借鉴思考

机理建模



优化模型



优化求解



敏感性分析



模型迁移

- 建模的原则还是要“一切有理论支撑”，不要轻信ai列出的式子。如果没有很好的依据则考虑简化表达式，简化不意味着不好。

学术文献查找：

Google Scholar : <https://scholar.google.com/>

arXiv : <https://arxiv.org/>

中国知网 : <https://www.cnki.net/>

- 建议可以提前熟悉PowerPoint, Photoshop, Adobe Illustrator等作图软件（以及DrawIO、Visio等作图网站），尝试去复现一些顶奖论文的示意图绘制，了解一些常见的功能。这些技能在以后的学习及科研中都有很大的帮助





上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

谢谢观看

祝同学们取得好成绩

联系方式: sjtu.yiyang.jin@gmail.com